THE DISCREPANCY OF SYLVESTER–HADAMARD MATRICES IN ODD DIMENSION

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ABSTRACT. Let H_k be the $2^k \times 2^k$ Sylvester–Hadamard matrix and let $\operatorname{disc}(H_k)$ be the minimum of $\|H_k x\|_\infty$ over sign vectors x. It is known that $\operatorname{disc}(H_k) = 2^{k/2}$ for even k, and it has been conjectured that $\operatorname{disc}(H_k) = 2^{(k+1)/2}$ for every odd k. We show that $\operatorname{disc}(H_k) = 2^k - 2\rho(1,k)$, where $\rho(1,k)$ is the covering radius of the first-order Reed–Muller code of length 2^k. Known values and bounds for this covering radius then show that the conjecture holds for $k \le 7$ and fails for every odd $k \ge 9$; in fact $\operatorname{disc}(H_k) \le \frac{7}{8}\, 2^{(k+1)/2}$ for every odd $k \ge 9$. Determining $\operatorname{disc}(H_k)$ for odd $k \ge 9$ is therefore equivalent to determining $\rho(1,k)$.

1. INTRODUCTION

For a real $m \times n$ matrix A the discrepancy is

$$\operatorname{disc}(A) = \min_{x \in \{-1,1\}^n} \|Ax\|_\infty.$$

Index the rows and columns of the Sylvester–Hadamard matrix H_k by $\mathbb{F}_2^k$, so that $(H_k)_{a,y} = (-1)^{a \cdot y}$. By Parseval every $x \in \{-1,1\}^{2^k}$ satisfies $\|H_k x\|_2^2 = 2^{2k}$, hence $\|H_k x\|_\infty \ge 2^{k/2}$, and $\operatorname{disc}(H_k) = 2^{k/2}$ when k is even. The problem list [1] records the odd case as open and states the following.

Conjecture ([1, Conjecture 13]). For every odd k, $\operatorname{disc}(H_k) = \sqrt{2}\,\sqrt{2^k} = 2^{(k+1)/2}$.

The value $2^{(k+1)/2}$ is what one obtains from H_{k-1} and a bent function, and the question is whether it can be improved. The answer is a statement about Boolean functions that is classical in coding theory. For $f \colon \mathbb{F}_2^k \to \mathbb{F}_2$ let

$$W_f(a) = \sum_{y \in \mathbb{F}_2^k} (-1)^{f(y) + a \cdot y}$$

be its Walsh transform, and let $\mathrm{NL}(f)$ be the Hamming distance from f to the set of affine functions. The covering radius of the first-order Reed–Muller code $RM(1,k)$, the code of affine functions, is $\rho(1,k) = \max_f \mathrm{NL}(f)$.

Theorem 1. *For every $k \ge 1$, $\operatorname{disc}(H_k) = 2^k - 2\rho(1,k)$.*

Corollary 2. (1) $\operatorname{disc}(H_k) = 2^{(k+1)/2}$ *for* $k \in \{1,3,5,7\}$.

(2) $\operatorname{disc}(H_9) \le 28 < 32 = 2^5$.

(3) *For every odd $k \ge 9$, $\operatorname{disc}(H_k) \le \frac{7}{8}\, 2^{(k+1)/2}$. In particular the Conjecture fails for every odd $k \ge 9$.*

(4) *For every odd $k \ge 15$, $\operatorname{disc}(H_k) \le \frac{27}{32}\, 2^{(k+1)/2}$.*

2. Proofs

Proof of Theorem 1. Every $x \in \{-1,1\}^{\mathbb{F}_2^k}$ is $x = (-1)^f$ for a unique $f\colon \mathbb{F}_2^k \to \mathbb{F}_2$, and then $(H_k x)_a = \sum_y (-1)^{a\cdot y + f(y)} = W_f(a)$. Writing $d(f,g)$ for Hamming distance,

$$d(f, a\cdot y) = \tfrac{1}{2}\big(2^k - W_f(a)\big), \qquad d(f, a\cdot y + 1) = \tfrac{1}{2}\big(2^k + W_f(a)\big),$$

so the distance from f to the affine functions is $\mathrm{NL}(f) = \frac{1}{2}\big(2^k - \max_a |W_f(a)|\big)$. Hence

$$\|H_k x\|_\infty = \max_a |W_f(a)| = 2^k - 2\,\mathrm{NL}(f),$$

and minimizing over x is maximizing $\mathrm{NL}(f)$ over f. □

Lemma 3. *Let $f\colon \mathbb{F}_2^k \to \mathbb{F}_2$ and let $b\colon \mathbb{F}_2^m \to \mathbb{F}_2$ be bent (m even). The function $g(y,z) = f(y) + b(z)$ on $\mathbb{F}_2^{k+m}$ satisfies $\max|W_g| = 2^{m/2}\max|W_f|$. Consequently $\mathrm{disc}(H_{k+m}) \le 2^{m/2}\,\mathrm{disc}(H_k)$ for every even $m \ge 0$.*

Proof. The sum defining $W_g(a,c)$ factors as $W_f(a)\,W_b(c)$, and $|W_b(c)| = 2^{m/2}$ for every c because b is bent. Bent functions exist in every even dimension [7]. The second statement follows from Theorem 1, applied with f of maximal nonlinearity. □

Proof of Corollary 2. (1) The covering radii are $\rho(1,1) = 0$ and $\rho(1,3) = 2$ by inspection, $\rho(1,5) = 12$ [2], and $\rho(1,7) = 56$ [3]. Theorem 1 gives $\mathrm{disc}(H_k) = 2, 4, 8, 16$ for $k = 1, 3, 5, 7$, which is $2^{(k+1)/2}$ in each case.

(2) Kavut and Yücel [6] found nine-variable Boolean functions of nonlinearity 242, so $\rho(1,9) \ge 242$ and $\mathrm{disc}(H_9) \le 512 - 484 = 28$. (Nonlinearity 241, found earlier by Kavut, Maitra and Yücel [5], already gives $\mathrm{disc}(H_9) \le 30$.)

(3) Every odd $k \ge 9$ is $9 + m$ with m even. By Lemma 3 and (2), $\mathrm{disc}(H_k) \le 28 \cdot 2^{m/2} = \frac{28}{32}\,2^{(9+m+1)/2}$.

(4) Patterson and Wiedemann [4] showed $\rho(1,15) \ge 16276$, so $\mathrm{disc}(H_{15}) \le 32768 - 32552 = 216 = \frac{27}{32}\cdot 256$. Lemma 3 carries this to every odd $k \ge 15$. □

3. Remarks

Remark 4. Part (4) uses only results available since 1983, so the Conjecture was false for $k \ge 15$ before it was stated. The first odd dimension in which it fails is $k = 9$.

Remark 5. By Theorem 1 the odd case of the problem in [1] is the problem of the covering radius of $RM(1,k)$ for odd k, open since the 1970s. For odd k the known bounds are

$$2^{k/2} < \mathrm{disc}(H_k) \le 2^{(k+1)/2},$$

where the strict lower bound is the classical fact that bent functions exist only in even dimension, and the upper bound is improved by the constructions above for every odd $k \ge 9$. Any new lower bound on $\rho(1,k)$, and any upper bound on $\rho(1,k)$, transfers to $\mathrm{disc}(H_k)$ through Theorem 1 without loss.

Remark 6. The list [1] also states Conjecture 12, $\limsup_{n\to\infty} \sup_{A\in\{\pm1\}^{n\times n}} \mathrm{disc}(A)/\sqrt{n} > 1$, and notes that Conjecture 13 would imply it through H_k, k odd. By Theorem 1, Sylvester–Hadamard matrices witness Conjecture 12 if and only if

$$\limsup_{k \text{ odd}} \big(2^k - 2\rho(1,k)\big)\, 2^{-k/2} > 1.$$

Corollary 2 bounds this lim sup by $\frac{27}{32}\sqrt{2} \approx 1.19$; whether it exceeds 1 is a question about the covering radius of $RM(1,k)$ that the present note leaves open.

Remark 7. The repository that accompanies this note contains `check.py`, which verifies Theorem 1 by exhaustive search for $k \leq 4$ and checks the product identity of Lemma 3 on random functions.

References

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